

Symptomatic disc herniation and serum lipid levels

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Insufficient blood supply to the intervertebral disc (IVD) has been proposed to play a role as causative factor in IVD degeneration. There is an association between IVD diseases and increased risk of dying of ischaemic heart disease. Obesity and tobacco are potential risk factors for degenerative IVD disease. High blood cholesterol and triglycerides serum levels are risk factors for atherosclerosis, and could be responsible for a decreased in the blood supply to the already poor vascularized IVD. We performed a frequency-matched case-control study to determine the serum levels of patients with symptomatic herniated lumbar disc. We examined the fasting serum lipid levels in 384 subjects who were operated at our institution. Group 1 included 169 consecutive patients (115 men and 54 women; mean age: 59.1 years, range 29–85) who underwent surgery for symptomatic disc herniation. Group 2 (control group) included 169 patients (115 men and 54 women; mean age: 61 years, range 26–86) who underwent arthroscopic meniscectomy for a meniscal tear in the same period. These patients were frequency-matched by age (within 3 years) and gender with patients of Group 1. Sera were extracted from blood samples and the concentrations of total cholesterol (TC) and triglycerides (TG) were determined. When comparing the two groups, patients with symptomatic herniated lumbar disc showed statistically significant higher triglyceride concentration ($P = 0.02$) and total cholesterol concentration ($P = 0.01$). Serum lipid levels may be a risk factor for IVD pathology. An enhanced understanding of these factors holds the promise of new approaches to the prevention and management of IVD pathology.